

Sep 8, 2024 (Due: 08:00 Sep 15, 2025)

1. The 32-bit floating-point format discussed in the lecture is the IEEE-754 single precision format consisting of 1 sign bit, 8 bits of exponent, and 23 bits of significand. Estimate the maximum finite floating-point number, the minimum positive (normal) floating-point number, as well as a tight upper bound on the relative representation error for the IEEE-754 single precision format.

What about the IEEE-754 double precision format consisting of 1 sign bit, 11 bits of exponent, and 52 bits of significand?

2. What is the hexadecimal representation of $3/7$ in IEEE single precision floating-point format? Explain how the number is encoded.

3. Let $U \in \mathbb{R}^{n \times n}$ be upper triangular and nonsingular. Provide two different implementations for solving the linear system $Ux = b$, where $b \in \mathbb{R}^n$ is a given vector.

4. Implement Gaussian elimination (without pivoting) for solving nonsingular linear systems. You may assume that no divide-by-zero error is encountered. Measure the execution time of your program in terms of matrix dimensions and visualize the result by a log-log scale plot. (You may generate your test matrices with normally distributed random elements.)

5. (optional) Write a program to evaluate the harmonic series using IEEE single precision floating-point numbers. What do you observe? Give an explanation.

6. (optional) Read the paper “What every computer scientist should know about floating-point arithmetic” by David Goldberg.